

PRAKALPA SATYAL

Computer Engineering Student

Gatthaghar, Bhaktapur | 9860199571 | praksatyal@gmail.com | github.com/Praaaaks | [LinkedIn](#)

PROFESSIONAL SUMMARY

Final-semester Computer Engineering student at Kathmandu University with hands-on experience building AI-integrated and full-stack applications across mobile, web, and simulation environments. Has delivered projects spanning real-time mobile platforms, NLP-based detection systems. Dedicated learner and passionate developer, always seeking new opportunities to build and collaborate.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, HTML5, CSS3

Frameworks & Libraries: React-Native, Node.js, PyTorch

Databases & Backend: Supabase, PostgreSQL, MongoDB

AI / ML: Reinforcement Learning, Natural Language Processing (NLP)

Tools & Platforms: Git, GitHub, Postman

Spoken Languages: Nepali, English

PROJECTS

Sahayatri — University Ride-Sharing Platform

React Native, TypeScript, Supabase, PostgreSQL

- Designed a normalized Supabase (PostgreSQL) database schema managing users, rides, notifications, messages
- Architected a live messaging system leveraging Supabase Realtime (Postgres subscriptions) for low-latency driver-passenger communication.

D-Brief — Decentralized Disaster Intelligence and Verification Platform

FastAPI, Python, REST APIs, SQLAlchemy

- Developed a scalable RESTful API with FastAPI and SQLAlchemy to process real-time disaster reports
- Orchestrated an end-to-end AI pipeline

GuardMeet — Multilingual Hate Speech Detection

JavaScript, Chrome Extension API (Manifest V3), REST APIs

- Collaborated on a real-time, multilingual (English and Nepali) hate speech and profanity detection system for virtual meetings and classrooms, powered by XLM-RoBERTa and OpenAI Whisper.
- Created real-time DOM inspection & alerting UI to notify meeting participants of harmful content.
- Built a streaming audio processing pipeline to capture, encode, and transmit 4-second chunks to an AI inference backend.

EDUCATION

Kathmandu University — B.E. in Computer Engineering

Dhulikhel, Kavrepalanchowk | Final Semester

Dec 2022 – Present, Expected 2027

HACKATHONS & ACHIEVEMENTS

- IT Meet Hackfest 2025
- Google Trilingual Machine Translation (TMT) Hackathon 2026 — Honorable Mention, organized by the Information and Language Processing Research Lab (Dept. of CSE, Kathmandu University) in association with Google Research.